SOLUÇÕES FICHA 1

NOÇÕES BÁSICAS DE CÁLCULO

1.1.

a) $-\frac{\sqrt{3}}{2}$ **b)** $\frac{\sqrt{3}}{2}$ **c)** -1 **d)** $\frac{2\sqrt{3}}{3}$

e) $-\frac{\sqrt{3}}{3}$ **f)** -2 **g)** $-\frac{\pi}{6}$ **h)** $\frac{2\pi}{3}$

i) $\frac{\pi}{4}$ **j)** $\frac{\sqrt{3}}{3}$ **k)** $\frac{\sqrt{2}}{2}$ **l)** 1

m) 4 **n)** 4 **o)** -2 **p)** -2

1.2.

a) $t = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}$

b) $x = -\frac{15\pi}{4}, -\frac{9\pi}{4}, \frac{9\pi}{4}, \frac{15\pi}{4}$

c) $x = -\frac{3\pi}{8}, \frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}$

d) Sem solução

e) $x = \frac{7\pi}{6} + \frac{7\pi n}{3}, n = 0, \pm 1, \pm 2, \ldots$

f) $x = \frac{3}{2};\ 9{,}0013 + \frac{4\pi}{3}n;\ 3{,}2417 + \frac{4\pi}{3}n$, $n = 0, 1, 2, \ldots$

1.3.

a) $f^{-1}(x) = \frac{1}{6}(x - 15)$

b) $f^{-1}(x) = \sqrt[3]{x - 6}$

c) $f^{-1}(x) = \frac{1}{5}(x^7 - 8)$

d) $f^{-1}(x) = \frac{4x - 1}{9 + x}$

1.4.

a) $\frac{\pi}{3}$

b) $D_f = [-2, 0] \quad D'_f = \left[-\frac{5\pi}{3}, \frac{\pi}{3}\right]$

c) $x = \frac{\sqrt{3} - 2}{2}$

d) $f^{-1}(x) = \cos\left(\frac{\pi}{6} - \frac{x}{2}\right) - 1$

e) $S = \left[-2, -\frac{1}{2}\right]$

1.5.

a) $\frac{\pi}{3}$

b) $D_f = \mathbb{R} \setminus \{-3\} \quad D'_f = \left[-\frac{\pi}{3}, \frac{2\pi}{3}\right] \setminus \left\{\frac{\pi}{6}\right\}$

c) $x = -\sqrt{3} - 3$

d) $f^{-1}(x) = \cot\left(x - \frac{\pi}{6}\right) - 3$

1.6.

a) 31

b) 3

1.7.

1a)	$-\frac{4}{9}$	**2a)**	0	**3a)**	$-\frac{\sqrt{2}}{2}$	**4a)**	$+\infty$	**5a)**	0
1b)	$-\frac{4}{9}$	**2b)**	0	**3b)**	$\frac{\sqrt{2}}{2}$	**4b)**	0	**5b)**	$+\infty$

1.8. Confirmar os esboços com uma calculadora gráfica.